

CONCORD, NH, USA

WMO: 726050

Lat: 43.205N

Lon: 71.503W

Elev: 105

StdP: 100.07

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14745

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -19.5	(c) -16.7	(d) -25.8	(e) 0.4	(f) -17.4	(g) -23.4	(h) 0.5	(i) -14.5	(j) 11.4	(k) -8.0	(l) 10.4	(m) -8.4	(n) 2.5	(o) 310	(p) 0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	32.3	22.0	30.6	21.1	29.1	20.3	23.7	29.3	22.8	27.9	21.9	26.5	4.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	17.0	25.8	21.2	16.1	25.0	20.4	15.2	24.3	71.2	29.2	67.6	28.0	64.5	26.5	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.4	8.3	7.4	-24.0	35.0	3.6	1.7	-26.6	36.2	-28.7	37.2	-30.7	38.1	-33.3	39.3
			-24.3	25.2	3.5	1.0	-26.8	26.0	-28.8	26.6	-30.8	27.2	-33.3	27.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	-5.6	-4.0	0.7	7.4	13.5	18.6	21.5	20.7	16.3	9.6	3.6	-2.0		
	DBStd	10.45	6.14	5.56	5.27	4.61	4.22	3.80	2.99	3.09	4.07	4.35	4.74	5.06		
	HDD10.0	1921	483	392	293	104	12	0	0	0	2	61	199	374		
	HDD18.3	3894	741	625	548	329	160	43	6	13	86	271	442	632		
	CDD10.0	1348	1	0	4	27	121	258	358	331	192	49	7	1		
	CDD18.3	279	0	0	0	1	11	51	106	84	26	1	0	0		
	CDH23.3	3020	0	0	8	42	210	577	1063	834	263	23	0	0		
	CDH26.7	996	0	0	2	12	66	195	377	268	73	2	0	0		
(14) Wind	WSAvg	2.6	2.9	3.2	3.3	3.1	2.7	2.4	2.2	1.9	2.0	2.3	2.5	2.7		
(15) Precipitation	PrecAvg	993	66	67	77	80	85	88	88	88	85	95	89	85		
	PrecMax	1475	206	228	180	183	282	225	192	271	240	370	191	191		
	PrecMin	614	10	1	22	21	13	16	27	0	11	23	13	20		
	PrecStd	189	36	37	34	35	49	47	42	48	49	61	39	38		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	14.1	22.7	28.3	31.8	33.7	34.2	33.9	31.8	26.2	19.7	15.0		
		MCWB	12.1	9.0	13.5	16.2	19.3	22.8	23.5	22.9	22.0	17.6	14.6	10.9		
	2%	DB	8.3	9.3	15.5	23.3	28.5	31.1	32.2	31.4	28.8	22.9	16.9	10.4		
		MCWB	5.7	5.1	9.0	13.4	17.6	21.1	22.4	21.7	20.5	16.3	12.9	7.8		
	5%	DB	4.9	6.2	11.5	19.6	25.4	28.9	30.5	29.6	26.3	19.9	14.2	7.6		
		MCWB	2.5	2.9	6.2	11.7	15.9	19.9	21.5	21.0	19.0	15.2	10.6	5.2		
	10%	DB	2.6	3.8	8.5	16.5	22.6	26.7	28.8	27.9	24.2	17.4	11.5	5.2		
		MCWB	0.6	1.2	4.4	9.3	14.4	18.5	20.6	20.1	17.9	13.3	8.3	3.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	10.3	14.5	17.7	20.8	24.3	25.1	24.6	23.4	20.1	16.7	12.1		
		MCDB	12.7	12.9	21.1	24.1	29.1	30.8	31.7	30.6	28.7	22.9	18.8	14.5		
	2%	WB	5.9	5.5	10.3	15.1	19.2	22.9	23.9	23.3	21.9	17.8	14.0	8.3		
		MCDB	7.7	8.3	14.4	21.2	25.4	28.7	29.5	28.4	26.3	20.9	16.1	9.8		
	5%	WB	2.9	3.2	6.9	12.7	17.6	21.7	22.9	22.4	20.7	15.7	11.2	5.5		
		MCDB	4.6	5.8	10.4	17.8	23.3	26.5	28.1	27.0	24.1	19.2	13.3	7.3		
	10%	WB	0.8	1.5	4.8	10.2	15.8	20.3	21.9	21.5	19.5	13.9	8.8	3.1		
		MCDB	2.1	3.4	8.2	15.8	20.8	24.4	26.6	25.8	22.8	16.6	11.1	4.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	10.7	10.8	12.9	13.3	12.5	12.6	12.9	13.1	12.3	10.4	9.3		
		MCDBR	11.5	13.2	16.0	19.2	18.4	15.6	14.8	14.9	15.5	16.1	14.1	12.3		
	5% WB	MCWBR	9.7	9.7	10.2	10.3	8.8	7.0	6.2	6.5	7.8	9.9	10.2	9.9		
		MCWBR	10.6	11.8	14.3	16.3	15.8	12.9	12.7	12.1	12.3	13.5	12.1	11.5		
(40) Clear-Sky Solar Irradiance	taub		0.281	0.293	0.319	0.354	0.390	0.433	0.454	0.422	0.370	0.333	0.308	0.286		
		taud	2.431	2.397	2.438	2.410	2.346	2.247	2.195	2.274	2.400	2.516	2.538	2.509		
	Ebn at Noon		862	910	926	913	883	842	818	835	857	854	826	826		
		Edh at Noon	74	91	100	111	122	136	141	126	103	81	68	64		
(44) All-Sky Solar Radiation	RadAvg		1.71	2.58	3.64	4.57	5.05	5.45	5.72	5.19	4.21	2.68	1.82	1.36		
	RadStd		0.12	0.20	0.23	0.50	0.60	0.52	0.38	0.28	0.27	0.26	0.17	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+80	N/A
(47) Regional (75 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+87	+34

Nomenclature: See separate page